



Detailed Schedule for Contributed Symposia for ICBF 2026

Date: July 13 Monday	Room: IRC 2 (503 seats)	
Time	Title	Presenter:
8:30	Welcome to ICBF 2026 and the plenary sessions	Don MacKinlay, DFO SEP, Colin Brauner, Co-Chairs,
8:45	American Fisheries Society, Physiology section "Award of Excellence" introduction	Erika Eliason AFS Physiology President
9:00	"Award of Excellence Lecture: Adaptations and acclimation to the environment in fishes: Integrating basic and applied research	Colin Brauner UBC Zoology
9:45	From shark gills to coral reefs: Discovering mechanisms and developing solutions	Martin Tresguerres UC San Diego
10:30 – 11:00	BREAK	
11:00	Using mechanistic physiology to inform salmon conservation	Erika Eliason, DFO Science
11:45	The importance of mechanistic physiology in addressing real-world problems	Chris Wood, UBC Zoology
12:30 – 14:00	LUNCH	
	STRESS I Stress in fish: A tribute to George Iwama	Matt Vijayan, and Paige Ackerman,
14:00	Everything is connected: Stress, fish health, and the lab that shaped my career	Ackerman, Paige
14:15	Science and socials - a decade of adventures with the Iwama lab	Morgan, John
14:30	It's complicated! Incorporating environmental complexity into multiple stressor experiments	Todgham, Anne
14:45	Stress in fish or de-stress in me: Lessons learned to identify which genes are most sensitive to chemical pollutants in trout alevins.	Basu, Niladri
15:00	Is scale cortisol the long-sought biomarker for chronic stress in fishes?	Bernier, Nick
15:15	Transcriptome responses in Atlantic salmon and Chinook salmon livers following microcystin-LR exposure	Ellingson, Brandon
15:30 - 1600	BREAK	
16:00	Beneficial effects of environmental stress in fish: Integrating stress physiology with aquaculture applications	Nakano, Toshiki
16:15	Out of air again: Effects of repeated hypoxic stress on the hypoxia tolerance of brook trout	Stairs, Chandler
16:30	Operationally simple environmental RNA collection using polyester wool enables accurate detection of physiological gene expression rhythms in cultured fish	Takemura, Akihiro
16:45	Different mucosae show differential physiological responses associated to specific stressors	Tort, Lluís

17:00	The ameliorating effect of calcium on alkaline stress in fish. Claudin 14 takes the stage.	Wilson, Jonathan
17:15	Reminiscing George Iwama's influence: Extracellular heat shock protein 70 modulate stress signaling in rainbow trout	Vijayan, Matt
17:30	The physiological effects of metals: finding peace in the study of stress	McGeer, Jim
17:45	END	

Date: July 13 Monday	Room: IRC 6 (181 seats)	
Time	Title	Author(s)
12:30 – 14:00	LUNCH	
	GROWTH Current trends in growth and metabolism of fishes	Brian Small, Brian Peterson and Suraj Unniappan
14:00	How marine three-spined stickleback modulate metabolic pathways in chronic hypoxia	Negrete Jr., Benjamin
14:15	Growth reduction and aerobic metabolic changes induced by warming and hypoxia in Gulf of St. Lawrence redfish (<i>Sebastes fasciatus</i>)	Guitard, Joëlle
14:30	Towards the sustainable production of more robust and resilient European sea bass juveniles in a climate change scenario.	Gutierrez, Joaquin
14:45	Routine metabolic rate across early developmental stages of Atlantic salmon (<i>Salmo salar</i>) in relation to survival and growth	Bair, Halli
15:00	Warm and well-fed: how ration availability shapes aerobic scope and growth across temperatures in juvenile Chinook salmon	Zillig, Kenneth
15:15	Growth inhibition under high-density and brain transcriptome analysis in rainbow trout	Fujishiro, Kouyou
15:30 - 1600	BREAK	
16:00	Inflammation modulates central regulation of homeostasis in rainbow trout	Soengas, José L
16:15	Baseline electrogenic characterization of methionine transport and gut barrier function, and responses to specific anti-nutritional factors in Atlantic salmon (<i>Salmo salar</i>) and rainbow trout (<i>Oncorhynchus mykiss</i>)	Subramaniam, Marina
16:30	Central activation of chaperone-mediated autophagy reduces appetite by fine-tuning hypothalamic amino acid pools: new insights from fish	Reji, Steffi
16:45	Do household plasticizers affect taste receptors in extra-oral tissues of female zebrafish?	Naher, Nazmun

17:00	Crosstalk between the gut-brain axis and the endocannabinoid system in rainbow trout: Implications for appetite regulation	Blanco, Ayelén M.
17:15	Artificial light including blue-green wavelengths promote growth of <i>Rachycentron canadum</i> with mRNA upregulation of appetite, growth axis and lipid synthesis	Zhu, Yafan
17:30	Chronic high-fat diet intake disrupts gonadal growth and development in zebrafish	Mustapha, Umar Farouk
17:45	END	

Date: July 13 Monday	Room: IRC 1 (120 seats)	
Time	Title	Author(s)
12:30 – 14:00	LUNCH	
	CHANGE II Investigating the interrelated responses of climate, contaminants, & biodiversity loss in fishes	Tamzin Blewett and Ivan Cadonic
14:00	Environmental variability in estuaries modulates pfas exposure and early life stage development in fishes	Ackerly, Kerri Lynn
14:15	Physiological and molecular impacts of Nickel exposure during thermal acclimation in Rainbow trout	Cadonic, Ivan
14:30	Beyond bottom-up effects: suspended sediment directly suppress reef fish assemblages and herbivore functions	Schakmann, Mathias
14:45	Glacial retreat as a vector for mercury (Hg) exposure: Evidence for uptake of glacially derived Hg in a coastal food web	Nielsen, Kristin
15:00	Copper and a common cold: Understanding the immune system modulation of physiological & behavioural impacts of copper exposure in goldsinny wrasse.	Martin, Sidney
15:15	Will wildfire ash smoke the salmon? Disentangling acid-base and contaminant impacts in UK and Australian fishes	Kwan, Garfield
15:30 - 1600	BREAK	
16:00	Characteristics of drought in Amazonian-river systems: relevance to fish communities.	Johannsson, Ora
16:15	Climate vs. weather: short-term environmental change leads to altered phenotypes across generations	Martinez Bautista, Gil
16:30	Individual and cumulative effects of nylon microplastic fibres and warming temperature on threespine stickleback behavior	Hajji, Angelina Lyla
16:45	Caught in a thermal loop: Unraveling the molecular, biochemical, and whole-animal responses to diel thermal variability in zebrafish	Clow, Tanner

17:00	Stress in the waters: Physiological and morphological responses to environmental stressors in imperiled fishes from Southwestern Ontario	Dodsworth, Emily
17:15	END	

Date: July 13 Monday	Room: IRC 4 (120 seats)	
Time	Title	Author(s)
12:30 – 14:00	LUNCH	
	ANCIENT FISHES Ancient fishes in comparative physiology: Agnathan and chondrichthyan fishes	Gary Anderson, Ian Bouyoucos and Alyssa Weinrauch
14:00	Evidence for cortisol in a shark	Weinrauch, Alyssa
14:15	The latest research on spiny dogfish aquaporin gene/protein expression...	Cutler, Christopher
14:30	Dietary state influences olfactory and cardiovascular responses to odorants in North Pacific Spiny Dogfish	Edwards, Tyler
14:45	Impacts of warming on cardiac function in the temperate Pacific spiny dogfish	Folkerts, Erik
15:00	Molecular characterization of potassium channels and their osmoregulatory role in the gills of sea lamprey	Cannizzaro, Ava
15:15	Anoctamin1/TMEM16A expression in the spiny dogfish: Is one of its roles likely to be osmoregulatory?	Cutler, Christopher
15:30 - 1600	BREAK	
16:00	Effects of salinity transfer on cardiac function in the partially euryhaline North Pacific spiny dogfish	Hutton, Melody
16:15	Blueprints for survival: Genetic mechanisms underpinning physiological performance in the epaulette shark <i>Hemiscyllium ocellatum</i>	Hasenei, Aaron
16:30	An integrative approach toward quantifying physiological responses to stressors in spiny dogfish	Debaere, Shamil
16:45	How pacific hagfish survive reoxygenation following extended anoxia	Yutsyschyna, Maria
17:00	Physiological performance and stress resilience in larval sea lamprey	Wilkie, Michael
17:15	Regulatory mechanisms of selective follicle development, maturation and ovulation leading to an ovarian hierarchy in cloudy catshark	Inoue, Ryotaro
17:30	END	

Date: July 13 Monday	Room: IRC 5 (120 seats)	
Time	Title	Author(s)
12:30 – 14:00	LUNCH	
	MOTION I Metabolism in motion: Mechanics, physiology, and ecology of fish movement	Katja Anttila (Yangfan Zhang, and George Lauder)
14:00	Linking energetics, acoustic telemetry, hydrodynamics, and neurophysiology to understand wild fish behavior	Liao, James
14:15	An optimised workflow for automated multiple fish tracking in challenging experimental settings	Bellio, Martina
14:30	The effect of salinity on the hypoxia tolerance, metabolism and swimming performance of a euryhaline teleost, white perch	Nelson, Jay
14:45	Survival of the slowest? Using activity to predict climate change vulnerability of fishes	Dichiera, Angelina
15:00	Exercise training in rainbow trout: from cardiac action potential modulation to heart rate adjustments	Le, Thuy
15:15	Kinematic modulation for flow exploitation as a foundation for Locomotor energy conservation	Zhang, Yangfan
15:30 - 1600	BREAK	
	MOTION II Physiology in motion: Linking movement, metabolism, and life history of fishes	Christian Bihun and Erin Stewart
16:00	Exploring the ecological relevance of metabolism and bioenergetics for wild fishes: new technological approaches and key questions	Brownscombe, Jacob
16:15	(Continued)	
16:30	Quantifying burst swimming in sockeye salmon during their 1100km upstream migration	Birnie-Gauvin, Kim
16:45	Brook trout in hot (and hypoxic) water	Reeve, Connor
17:00	The effect of size, age and sexual maturity on acoustic tagging success in an annual, endangered fish species	Steel, Anna
17:15	Telemetry tracking and physiological sampling indicate large smolt survival differences at a fish-friendly flood pump	Lingard, Stephanie
17:30	Evaluating the "fish-friendliness" of flood pumps using Sensor Fish	Kinnard, Hollis
17:45	Investigating walleye activity with accelerometer-bearing acoustic tags in the Great Lakes	Bihun, Christian
18:00	END	

Date: July 14 Tuesday	Room: IRC 2	
Time	Title	Author(s)
	STRESS II Stress in fish I (regulation of HPI) and II (stress effects on physiology)	Luis Tort, Matt Vijayan, Nicholas Bernier and Katie Gilmour
9:00	The effects of PFAS on the stress axis in zebrafish	Gorissen, Marnix
9:15	Early life glucocorticoid overexposure primes cell-type specific stress hypersensitivity in adult zebrafish	Choi, Min-Kyeong
9:30	Interactive effects of thermal and social stress on developmental stress physiology in zebrafish	Vámos, Sofia
9:45	Nucleobindin-derived peptides cause stress in male zebrafish	Sethi, Paalki
10:00	Brain drain: applying dynamic energy budget theory to the regulation of adult neurogenesis	Alderman, Sarah
10:15	Cortisol alters the hypothalamic lipidome in rainbow trout	Antomagesh, F.
10:30 – 11:00	BREAK	
11:00	Investigating cortisol as a mediator of acute thermal tolerance in the Atlantic killifish	Flatt, Emma
11:15	Cellular thermal profiles in the mucus of wild bull trout	Lazaro-Côté, Analisa
11:30	Transcriptional profiling provides insights into how fasting affects the thermal tolerance and physiological response to acute thermal stress in lake sturgeon	Bernier, Nick
11:45	miR-140-5p as a biomarker of thermal challenges in the European seabass	Sniecinski, Elisa
12:00	Using transcriptomics to explore variation in cold stress physiology and resilience of gizzard shad to cold snaps	Waraniak, Justin
12:15		
12:30 – 14:00	LUNCH	
14:00	Leptin function in hypoxia and its implications in the adaptive stress response in tilapia	Borski, Russell
14:15	Stress endpoints in hatchery-raised striped mullet, in response to salinity challenge.	Merlo, Reilly
14:30	Sub-lethal impacts of sea lamprey parasitism on the stress responses and energy reserves of rainbow trout	Czuba, Karolina
14:45	Social history and stress adaptation: oxidative stress and mitochondrial adaptation in cichlid fish	Dijkstra, Peter
15:00	Modulation of stress responses through the diet: a natural additive to improve gilthead seabream (<i>Sparus aurata</i>) resilience during transport	Cartan, Sara
15:15	The air breathing physiology and temperature and hypoxia tolerance of the oriental weatherloach, (<i>Misgurnus anguillicaudatus</i>)	Burbules, Katherine
15:30	END	
15:30 – 19:30	POSTER SESSION	

Date: July 14 Tuesday	Room: IRC 6	
Time	Title	Author(s)
9:00	NEURO Fish integrative neurobiology in changing environments: Introduction	Martin Tresguerres, Mar Huertas and Trevor Hamilton
9:15	Resilient brains: behavioral and physiological homeostasis in highly variable environments	Hofmann, Hans
9:30	(Continued)	Hofmann, Hans
9:45	Mapping zebrafish neural activity following exposure to increased CO ₂ and temperature	Cordle, Jonathan
10:00	Characterization of environmental nitrite transport into the fish olfactory-brain axis	Restrepo, Nicole
10:15	Exposure to three levels of CO ₂ -induced freshwater acidification results in multi-tissue transcriptional and behavioural responses in medaka	Hasler, Caleb
10:30 – 11:00	BREAK	
11:00	Effects of environmental changes on the fish sense of smell: infections, pollution, and other horror stories.	Huertas, Mar
11:15	Changing environments affects neural nitric oxide pathways in fish	Moretta Urdiales, Maria Del Mar
11:30	Thermal modulation of sensory-motor integration: developmental hyperthermia and chemical alarm cues in larval zebrafish (<i>Danio rerio</i>)	Vámos, Sofia
11:45	Rising to a challenge: Identifying the mechanisms mediating ‘surfacing’ behaviour in zebrafish	Gilmour, Kathleen
12:00	The magic of mushrooms: Psilocybin influences behaviour in the mangrove rivulus, <i>Kryptolebias marmoratus</i>	Forsyth, Dayna
12:15	Trait stability in the open field test predicts selectively bred anxiety-like phenotypes in zebrafish	Johnson, Andréa
12:30 – 14:00	LUNCH	
14:00	Climate-driven selection on neurobiological traits: insights from population genomics	Rennison, Diana
14:15	(Continued)	Rennison, Diana
14:30	Neurogenetic mechanisms of behavioral resilience and susceptibility in larval fish within changing environments	Tatarsky, Rose
14:45	Neurobiological responses to thermal pressure in natural threespine stickleback populations	White, Matthew
15:00	Structure-function relationships of the rainbow trout visual system following environmental hypoxia	McKinley, Abigail
15:15	Teleost cerebrospinal fluid response to future ocean conditions	Kwan, Garfield
15:30 – 19:30	POSTER SESSION	

Date: July 14 Tuesday	Room: IRC 1	
Time	Title	Author(s)
	CHANGE I Integrating physiology and behaviour in a changing environment	Brad Howell, Caleb Hasler, Nann Fangue and Steve Cooke
9:00	Mechanistic conservation of wild fishes demands integration of animal physiology and behaviour	Cooke, Steven J.
9:15	Hot and bothered: behavioural and physiological responses to warming in freshwater fish	de Bonville, Jeremy
9:30	Does temperature variability change how we interpret climate resilience in a nocturnal reef shark?	Emmons, Sophia M.
9:45	Evaluating climate change resilience in wild white sturgeon populations	Earhart, Madison
10:00	Does tolerance to fast warming predict tolerance to slow warming in fishes?	Lechner, Emily
10:15	Physiological and genomic evidence of differential thermal performance in Kokanee salmon (<i>Oncorhynchus nerka</i>)	Mayer, Noa
10:30 – 11:00	BREAK	
11:00	Measuring thermal tolerance and metabolic recovery times of wild salmonids after high thermal exposures: Implications for angling policies in Atlantic Canadian river systems	Morash, Andrea
11:15	Habitat thermal variability predicts upper thermal tolerance in a coastal triplefin fish	Johnson, Sheri
11:30	Linking metabolism, physiological performance and oxygen transport in Baltic cod exposed to prolonged mild hypoxia	Husen, Magnus
11:45	No air, no action: cod hit the slow lane under hypoxia	Birnie-Gauvin, Kim
12:00	Developmental neurobehavioural toxicity of Bisphenol F in zebrafish larvae	Hasan, A K M Munzurul
12:15	Oxygen supply and demand in diel vertically migrating mesopelagic fishes (Myctophidae)	Seibel, Brad
12:30 – 14:00	LUNCH	
14:00	Biological responses of juvenile brook trout (<i>Salvelinus fontinalis</i>) to the interactive effects carbon dioxide and temperature	Bellinger, Imogen
14:15	CO ₂ induced spatial working memory impairment and recovery in medaka (<i>Oryzias latipes</i>)	Hryciw-Pang, Sage
14:30	Changing temperatures: can cool water recovery offset high temperatures for partially armored threespine stickleback native to California?	Cooper, Cassidy
14:45	Effects of live-well chilling on the physiology and behaviour of smallmouth bass	Thapar, Myra
15:00	END	
15:30 – 19:30	POSTER SESSION	

Date: July 14 Tuesday	Room: IRC 4	
Time	Title	Author(s)
	TOXICOLOGY Fish talks on phys tox	Sarah Alderman and Todd Gillis
9:00	The impacts of dietary PFOA on the reproductive behavior of male zebrafish (<i>Danio rerio</i>)	Khursigara, Alexis
9:15	Assessing lithium toxicity across a salinity gradient in coastal marine fishes	Esbaugh, Andrew
9:30	Timing is everything: Critical windows of road salt toxicity in early life stage Pacific salmonids	Kilgour, Clare L.
9:45	Impacts of road salt exposure on water hardening in coho salmon eggs (<i>Oncorhynchus kisutch</i>)	Hall, Jack
10:00	Caffeine as an emerging contaminant: behavioral and physiological effects of acute exposure in zebrafish	Chaves Morera, Victoria
10:15	Effects of chronic nitrite exposure on reproduction and maternal transfer in a live-bearing fish model (<i>Xiphophorus couchianus</i>)	Parkman, Trinity
10:30 – 11:00	BREAK	
11:00	The common pesticide azoxystrobin is toxic to the fish heart	Ruhr, Ilan
11:15	Alterations in transepithelial potential and ionoregulation in goldfish following exposure to differentially degraded microplastics in two different water chemistries	Zink, Lauren
11:30	Signals in the stream: Field-to-lab evidence of molecular and biochemical stress in fish exposed to wastewater-borne drugs of abuse	Craig, Paul
11:45	Thyroid receptor activity reporter: modelling the fundamental mechanisms of thyroid disrupting pollutants in a changing climate	Simm, Hayden
12:00	Environmental fate meets fish function: rethinking risk of sea lice chemotherapeutants in coastal salmon habitat	Kennedy, Chris
12:15	Acute impacts of wildfire ash exposure on larval Pacific coho salmon (<i>Oncorhynchus kisutch</i>) physiology and survivorship	Munro, Cole
12:30 – 14:00	LUNCH	
14:00	Cardiorespiratory alterations and arrhythmias after acute exposure to water-accommodated fraction from crude oil in juveniles of two North American fish species	Weber, Lynn
14:15	The response of Pacific salmon to diluted bitumen, working towards a lifecycle perspective	Gillis, Todd
14:30	Phenanthrene is a ubiquitous cardiac toxicant with exacerbated impacts on fish	Shiels, Holly
14:45	END	
15:30 – 19:30	POSTER SESSION	

Date: July 14 Tuesday	Room: IRC 5	
Time	Title	Author(s)
	DEVELOPMENT Vulnerability and adaptations of developmental stages to global change stressors	Jeremy Bonville, Zara-Louise Cowan and Patrice Pottier
9:00	Which fish life stage is most vulnerable to warming?	Cowan, Zara
9:15	Thermal sensitivity of metabolism across the lifespan of a fish	Clark, Timothy
9:30	Effects of temperature on three life stages of partially armored threespine stickleback	Cooper, Cassidy
9:45	Comparing thermal tolerance of adult and embryonic life stages: Is survival a good measure of tolerance?	Harman, Adam
10:00	Unifying heat tolerance assessments across fish life stages	Pottier, Patrice
10:15	Temperature-driven embryo vulnerability as a recruitment bottleneck for European smelt (<i>Osmerus eperlanus</i>) in a warming estuary	Harvey, Megan
10:30 – 11:00	BREAK	
11:00	Early life vulnerability of coregonids to warming: species and population-specific thermal tolerance across Great Lakes and inland lakes	Zillig, Kenneth
11:15	Can fish embryos acclimate to warming?	Tomás, Mafalda
11:30	Physiological effects of chronic elevated CO ₂ exposure from fertilization to yolk-sac absorption and the freshwater-to-seawater transition in pink salmon (<i>Oncorhynchus gorbuscha</i>).	Lee, Seungyeon
11:45	Assessing seaweed's potential to buffer ocean acidification in Pacific herring embryos	Chalifoux, Virginie
12:00	The effect of fluctuating hypoxia on the physiology of larval coho salmon	Osborne, Courtney
12:15	Impact of maternally transferred methylmercury on embryo development and neurobehavior in fathead minnows (<i>Pimephales promelas</i>)	Ortega-Rodriguez, Celeste
12:30 – 14:00	LUNCH	
14:00	Lasting effects of variation in developmental temperature on later-life hypoxia tolerance	Eisenberg, Rachel
14:15	Early-life stress history determines thermal tolerance in larval zebrafish	Vámos, Sofia
14:30	Plasticity of cellular and whole animal responses after embryonic thermal and hypoxic stress related to climate change and thermal pollution.	Manzon, Richard G.
14:45	END	
15:30 – 19:30	POSTER SESSION	

Date: Tuesday July 14	Room: IRC 3	
Time	Title	Author(s)
	IMMUNOLOGY Fish immunology in a changing environment: Implications for aquaculture and conservation	Mark Fast and Ensiyeh (Forough) Ghanizadeh-Kazerouni
9:00	Spatial transcriptomic analysis of gill regeneration in Atlantic salmon using Hi-Plex RNAscope	Ghanizadeh-Kazerouni, Ensiyeh (Forough)
9:15	Salmonid red blood cells act as sensors of stress and infection	Dahle, Maria
9:30	Cut it out! Functional genomics of sea lice resistance in CRISPR-edited Coho salmon	Lo, Javon
9:45	Effects of temperature on host immunity and disease severity in salmonid bacterial kidney disease	Wheatley, Taylor
10:00	Integrating Host Immunity and Environmental Surveillance to Manage PKD in Rainbow Trout Aquaculture	King Wolf, Lisa
10:15	The effect of chronic CO ₂ exposure on the growth, metabolic rate and immune response of juvenile Lake Whitefish (<i>Coregonus clupeaformis</i>)	Fajei, Eyesun
10:30 – 11:00	BREAK	
11:00	Effect of florfenicol and oxytetracycline on the expression of proinflammatory genes in SHK-1 cell line from <i>Salmo salar</i> : focus on histone acetylation	Morera, Francisco J.
11:15	END	
12:30 – 14:00	LUNCH	
	EVOLUTION Fish evolution in human-altered environments	Andrew Whitehead
14:00	Evolutionary rescue from climate warming - insights from two artificial selection experiments	Jutfelt, Fredrik
14:15	Making sense of a hot mess: How larval fish may evolve in a warming, but variable ocean	Johnson, Darren
14:30	The ecology of gene expression in response to temperature in threespine stickleback	Rogers, Sean
14:45	The presence of domesticated alleles reduces survival and influences the phenotypes of rainbow trout in a semi-natural environment.	Schulte, Patricia
15:00	Evolved transcriptomic responses to multiple pollutant classes in Atlantic killifish	Rios, Carolina
15:15	Evidence of evolutionary adaptation following a series of environmental and demographic disasters in Pacific herring	Whitehead, Andrew
15:30	END	
15:30 – 19:30	POSTER SESSION	

Date: Wednesday July 15	Room: IRC 2	
Time	Title	Author(s)
	IONS I Ion and acid-base regulation in fishes	Greg Goss, Colin Brauner and Steve McCormick
9:00	Differential gene expression analysis of zebrafish gills from fish exposed to low pH, low ionic strength waters	Goss, Greg
9:15	What are the ecophysiological implications of periodic acidification of riverine systems?	Zimmer, Alex
9:30	Gill methylome shifts across multigenerational acidification in marine medaka	Tseng, Yung-Che
9:45	Linking molecular transport processes to acid–base regulation strategies in fishes	Shartau, Ryan
10:00	Distinct acid-base regulatory strategies in teleosts and elasmobranchs under ocean acidification	Chuang, Hsin-Ju
10:15	Characterizing bicarbonate transporters and their osmoregulatory roles in the Sea Lamprey intestine	Flowers, Dillon
10:30 – 11:00	BREAK	
11:00	Up-regulation of Fgf23 contributes to kidney stone formation in <i>stc1a</i> ^{-/-} mutant fish	Aburatani, Naotaka
11:15	Gill remodelling and increased Rh glycoprotein abundance enhance postprandial ammonia excretion by blood-feeding parasitic juvenile sea lamprey	Wilkie, Michael
11:30	Possible ammonia removal mechanism in the seahorse brood pouch revealed by rhesus glycoprotein expression and localization	Kawaguchi, Mari
11:45	Integrating gene expression, eQTL, and population genomic variation to reveal molecular adaptation in prickly sculpin (<i>Cottus asper</i>) across British Columbia	Liu, Shuang
12:00	Salinity and sex dependent ionoregulatory mechanisms in reproductively active <i>Fundulus heteroclitus</i>	Diakanastasis, Eleni
12:15	Hormonal control of determinants of hydromineral balance in gill and intestine of Atlantic killifish	Tipsmark, Christian
12:30 – 14:00	LUNCH	
14:00	Prolactin, cortisol, and extracellular osmolality regulate <i>cfr</i> , <i>ostf1</i> , and <i>sgk1</i> in tilapia ionocytes	Breves, Jason
14:15	Functional organization of ion transport in the trunk skin of European seabass (<i>Dicentrarchus labrax</i>): Evidence of an apical localization of Na ⁺ /K ⁺ -ATPase	Sundh, Henrik
14:30	Photoperiod- and salinity-dependent transcriptional regulation of renal CaSR and calcium transporters supports calcium homeostasis during seawater acclimation in Atlantic salmon	Takvam, Marius

14:45	Impact of commercial feed mineral load on urine composition and transcriptional regulation of divalent ion transporters in Atlantic salmon kidney	Denker, Elsa
15:00	The calcium binding protein S100a11 regulates zebrafish ionocyte quiescence–proliferation balance	Shih, Shang-Wu
15:15	END	
15:30	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Wednesday July 15	Room: IRC 6	
Time	Title	Author(s)
	AQUACULTURE Applying knowledge of fish biology to sustainable aquaculture	Andrea Frommel and Tillmann Benfey
9:00	Impact of intergenerational enrichment on juvenile westslope cutthroat trout methylome and phenotypic traits	Firth, Britney
9:15	Balancing natural and artificial selection in captive rearing programs	Maynard, Desmond
9:30	Comparative analysis of skin and gut microbiome dynamics in wild versus farmed Atlantic salmon smolts	Tang, Patrik Anthony
9:45	Developments in fish swimming technology for high-volume aquaculture systems	Herbert, Neill
10:00	Development of a gut dysbiosis assay for chinook salmon (<i>Oncorhynchus tshawytscha</i>)	Vandersteen, Wendy
10:15	Joint morphological and DNA metabarcoding gut content analysis reveals adult body size but not biological sex as factors influencing dietary composition divergence in multiple populations of cunner wrasse (<i>Tautoglabrus adspersus</i>).	Bender, Chris
10:30 – 11:00	BREAK	
11:00	Farmed sea lettuce (<i>Ulva fenestrata</i>) as a promising ingredient for rainbow trout (<i>Oncorhynchus mykiss</i>)	Andersson, Marica
11:15	Dietary and environmental interventions to optimize growth, stress, and reproductive outcomes in cultured longfin yellowtail, <i>Seriola rivoliana</i>	Chang, Ryan
11:30	Fasting reduces intestinal tissue mass and transport capacity in salmonids	de Fonseca, Raneesha
11:45	Use of alternative insect feed ingredients in sustainable wolf eel aquaculture	Dykstra, Jeremy
12:00	Enhancing growth, coloration, and fitness of koi (<i>Cyprinus rubrofuscus</i>) with purple bacteria	Opinion, April Grace
12:15	Insect-based diets modulate welfare, health, and gut microbiome functions in farmed Atlantic salmon	Tang, Patrik Anthony
12:30 – 14:00	LUNCH	

14:00	Effects of exercise training on cardiac morphology of rainbow trout	Anttila, Katja
14:15	Osmorespiratory compromise in triploid (sterile) salmonids	Benfey, Tillmann
14:30	Temperature and ploidy effect on red blood cell physiology in brook charr (<i>Salvelinus fontinalis</i>)	Virag, Victoria
14:45	Effect of environmental variation on the regulation of skin barrier function in rainbow trout in aquaculture	von Ehr, Michelle
15:00	Nephrocalcinosis in a jar: a multi-factor in vitro model of CO ₂ induced nephrocalcinosis	Emadi, Cameron
15:15	Set in stone: The effect of diet and carbon dioxide on nephrocalcinosis development	Mullins, Julia
15:30	Microbiome effects of fulvic and humic acid diets and baths	Niemann, Benjamin
15:45	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Wednesday July 15	Room: IRC 1	
Time	Title	Author(s)
	WELFARE Using behaviour and physiology to understand fish welfare	Katherine Sloman and Lynne Sneddon
9:00	Melanin-based skin pigmentation as an indicator of stress and welfare in Atlantic salmon (<i>Salmo salar</i>)	Gunnes, Mikkel
9:15	A fluttering heart: Use of biologgers during behavioural stress tests in rainbow trout	Sneddon, Lynne
9:30	From signals to states: biollogger-based classification of seabass welfare states in sea-cages	Hoyo-Alvarez, Esther
9:45	Behavioural and neurophysiological evidence for exercise choice as a welfare intervention in salmonids (<i>Salmo salar</i> and <i>Oncorhynchus mykiss</i>)	Bellio, Martina
10:00	Enrichment and aggression in fishes: implications for animal welfare	Dijkstra, Peter
10:15	Why “stress” no longer serves us: retiring an ambiguous term to advance fish welfare	Currie, Suzie
10:30 – 11:00	BREAK	
11:00	Effect of fishing stress on reproductive output of wild cod	Jacson, Lea
11:15	Turbine design shapes fish welfare: acute injury, mortality, and sublethal growth effects in juvenile white sturgeon	Zillig, Kenneth
11:30	Time-dependent effects of structural environmental enrichment on cognition and stress responses on juvenile European seabass	Rodríguez López, Noelia
11:45	Bringing together physiology, behavior, and cognition to understand the effect of environmental enrichment on positive welfare of farmed fish	Pettinau, Luca

12:00	The key importance of behavior and physiology to better understand fish welfare and resilience in aquaculture scenarios	Gesto, Manuel
12:15	Disruption of the endocannabinoid system during gastrulation produces long-term behavioral and reproductive deficits in zebrafish.	Khan, MD Hasanuzzaman
12:30 – 14:00	LUNCH	
14:00	Density and aggression in zebrafish: Resource Defense Theory explains conflicting results	Oldfield, Ronald
14:15	Improving the stress response and welfare of Nile tilapia, <i>Oreochromis niloticus</i> (Linnaeus, 1758) through environmental enrichment	Campang, Joseph
14:30	From behavior to biomass: early enrichment enhances trout welfare and performance	Jonassen, Ane
14:45	Enrichment, emotions and fish welfare - integrating behavioural measures to provide better fish care	Rose, Paul
15:00	Stress coping, allostasis, and the welfare of fish in aquaculture	Øverli, Øyvind
15:15	Improving fish welfare in the ornamental trade	Sloman, Katherine
15:30	Stunning challenges: Operational indicators flag failures, but neurological validation is needed to confirm stunning effectiveness in seabass and seabream	Saraiva, Joao
15:45	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Wednesday July 15	Room: IRC 4	
Time	Title	Author(s)
9:00	REPRODUCTION From reproductive physiology to molecular physiological ecology of fish: A tribute symposium to Rick Goetz	Josep Planas, Julien Bobe and Simon MacKenzie
9:15	Assessing heritable phenotypic variation in lake charr ecomorphs of Lake Superior	Goetz, Rick
9:30	Linking molecular responses to population consequences: differential effects of sea lamprey parasitism in siscowet and lean lake trout	Murphy, Cheryl
9:45	Gene presence-absence variation and DNA methylation landscapes underlying ecotype divergence in lake trout	Roberts, Steven
10:00	Phenotypic and genotypic variability of Lake Char sharing the same spawning reef complex	Euclide, Peter
10:15	Genomic evidence supporting panmixia of Pacific halibut in the Northeast Pacific Ocean	Jasonowicz, Andy
10:30 – 11:00	BREAK	
11:00	The influence of sea lamprey parasitism on Lake Charr reproductive physiology	Firkus, Tyler

11:15	Prostaglandin F2a is metabolized in taxon-specific manners demonstrating that this lipid may commonly serve as a precursor for species-specific spawning pheromones in ovulated fishes	Sorensen, Peter
11:30	Circulating microRNAs reveal egg-brain crosstalk and a brain-specific microRNA linked to the onset of the next reproductive cycle in iteroparous salmonids	Bobe, Julien
11:45	Mediating role of thyroid hormone in the regulation of temperature-induced changes in early gonadal differentiation in zebrafish	Tailor, Nurulain
12:00	The integrated role of arginine vasotocin in zebrafish spermatogenesis	Habibi, Hamid R
12:15	The reproductive biology of female Pacific halibut, a key species in the North Pacific Ocean	Planas, Josep V.
12:30 – 14:00	LUNCH	
14:00	A tale of rainbow trout myostatin – an unconventional journey	Biga, Peggy
14:15	From bacterial lipopolysaccharides to behavioural fever in fish; 3 decades of impactful collaborative research with Rick Goetz	MacKenzie, Simon
14:30	Closing remarks	Planas, Josep, Bobe, Julien, MacKenzie, Simon
15:00	END	
15:30	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Wednesday July 15	Room: IRC5	
Time	Title	Author(s)
	ADVANCES Recent advances in fish biology and physiology (general session)	Adam Harman and Ben Negrete Jr.
9:00	Get your hands wet! Developing high-impact laboratory exercises for undergraduate fish physiology courses.	Myrick, Christopher
9:15	Internal heating rates do not directly reflect water temperature in rainbow trout during CTmax trials	Sutherland, James
9:30	Growth, sexual maturation, and adult body size are linked with the ontogenetic development of aerobic metabolic scope in fish	Norin, Tommy
9:45	Open-source modular respirometry platform for measuring metabolic rate in fishes: PumpResp, SensResp, and SwimResp	Morozov, Sergey
10:00	Effects of warm exposure on swimming performance and aerobic metabolic capacity in juvenile Yukon River Chinook salmon	Snyder, Augustus

10:15	Changes to group dynamics reflect changes in individual behaviour under thermal stress in a model fish species	Tiddy, Izzy C
10:30 – 11:00	BREAK	
11:00	Does the rate of temperature change affect thermal plasticity in fishes?	Welsh, Emily
11:15	Modeling 125 years of spatial and temporal variability in climate and fish assemblages of temperate inland lakes	Lucas, Kelsey
11:30	Application of environmental RNA in the ecophysiological studies of fish	Wong, Marty Kwok-Shing
11:45	Impacts of body size and parasite load on the preferred temperature and avoidance temperatures of the diadromous fish <i>Galaxias maculatus</i>	Gradito, Maryane
12:00	The influence of food availability on metabolic thermal plasticity in overwintering brook char (<i>Salvelinus fontinalis</i>)	Hartenstein, Peyton
12:15	The impact of environmental conditions on Atlantic sturgeon (<i>Acipenser oxyrinchus oxyrinchus</i>) habitat use	Gonzalez, Kelly
12:30 – 14:00	LUNCH	
14:00	Oxygen uptake of rainbow trout during acute warming: effects of oxygen supersaturation and aerobic exercise	Bellinger, Imogen
14:15	Effects of experimental infections on host metabolism and oxidative damage in naïve and experienced populations of pumpkinseed sunfish (<i>Lepomis gibbosus</i>)	Mélançon, Vincent
14:30	Abiotic and biotic drivers of declining flounder abundance, distribution, and condition	Zavell, Max
14:45	END	
15:30	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Wednesday July 15	Room: IRC 3	
Time	Title	Author(s)
	RESPIRATION Respiratory gas exchange and circulation: The legacy of Dave Randall	Danielle McDonald and Alex Zimmer
9:00	Notes on my travels with Dave	Iwama, George
9:15	Insights from 'omics about anoxic crucian carp	Lefevre, Sjannie
9:30	Comparative red blood cell physiology: Uncovering novel cellular mechanisms to understand the impact of climate change on fishes	Harter, Till
9:45	Trading athleticism for hypoxia tolerance: Ontogenetic reorganisation of oxygen transport in coral reef fishes	Rummer, Jodie
10:00	Importance of blood (tissue) oxygen extraction to the environmental tolerances of fishes	Gamperl, Anthony
10:15	Evolution of respiratory tolerance to suspended sediment in freshwater fishes	Koseck, Benjamin
10:30 – 11:00	BREAK	

11:00	Respiratory physiology in the era of climate change	Esbaugh, Andrew
11:15	Interactive effects of temperature and deoxygenation on the acclimation capacity of Atlantic croaker	Hancock, Grace
11:30	Legacy effects of fishing on the climate resilience of fish	Goodrich, Harriet
11:45	To boldly go... to the surface: linking use of aquatic surface respiration to personality and hypoxia tolerance in zebrafish	Flear, Kaitlyn
12:00	When do plecospirans breathe air? developmental emergence and environmental drivers of aerial respiration in <i>ancistrus cirrhosis</i>	Dzialowski, Edward
12:15	Zebrafish biology: ignore sex at your peril	Burggren, Warren
12:30 – 14:00	LUNCH	
	POLAR The physiology, biology and ecology of polar fish	Pedro Guerreiro and Luis Vargas-Chacoff
14:00	Temperature sensing channels in Arctic gadids	York, Julia
14:15	Assessing fish community structure beneath the perennial ice-covered McMurdo Sound, Antarctica by environmental DNA metagenomics	Kim, Jinmu
14:30	Short-term resilience with hidden costs: physiological and molecular responses of juvenile <i>Notothenia rossii</i> to simulated marine heatwaves	Guerreiro, Pedro M
14:45	Photoperiod–temperature mismatch has impacts on feeding, metabolism, and thermal tolerance in the Antarctic spiny plunderfish <i>Harpagifer antarcticus</i>	Guerreiro, Pedro M
15:00	Impact of microplastic PVC and thermal stress on fish microbiomes and cellular stress genes in Antarctic and sub-Antarctic fish	Nualart Carrasco, Daniela Paz
15:15	Antarctic fish as a novel source for age-related disease therapeutics: KOPRI's bioprospecting initiative	Kim, Jin-Hyoung
15:30	END	
15:30	FREE AFTERNOON (MUSEUM EXCURSIONS)	

Date: Thursday July 16	Room: IRC 2	
Time	Title	Author(s)
9:00	IONS II Environmental physiology of life-history transitions: A tribute to Steve McCormick: Introduction	Ciaran Shaughnessy, Jill Leonard and Jason Breves
9:15	The light-pituitary TSH/brain DIO ₂ /T ₃ pathway for stimulating plasma GH and smoltification in juvenile Atlantic salmon	Björnsson, Björn Thrandur
9:30	Regulatory Plasticity of New Salmonid Na ⁺ /K ⁺ -ATPase α -subunit Paralogs and Functional Compensation in nka α 1b1/b2 Crispants	Nilsen, Tom Ole

9:45	Comparative transcriptomics of three Atlantic salmon (<i>Salmo salar</i>) stocks sampled at West Greenland	Ellingson, Brandon
10:00	The Intestinal “Saga” started with Steve! or What is happening in the intestine during the parr-smolt transformation	Sundell, Kristina Snuttan
10:15	Prolactin promotes putative pathways for Cl ⁻ absorption in Atlantic salmon ionocytes	Breves, Jason
10:30 – 11:00	BREAK	
11:00	Endocrine coordination of appetite and reproductive migration by dual ghrelin receptors in the Japanese eel	Tsukada, Takehiro
11:15	Stress and growth – is myostatin a link between these two?	Biga, Peggy
11:30	Metabolic reprogramming in zebrafish: an obesity model platform for bioactive extract bioprospecting.	Ferreira-Martins, Diogo
11:45	Thermal tolerance in chinook salmon varies with life-history and smoltification	Metzger, David
12:00	Acclimation to high salinity confers cross-tolerance to acute heat stress in <i>Fundulus heteroclitus</i>	Monette, Michelle
12:15	Photoperiod is controlling the Thyroid axis in the Antarctic fish <i>Harpagifer antarcticus</i>	Vargas-Chacoff, Luis
12:30 – 14:00	LUNCH	
14:00	Interrelations of the growth hormone/insulin-like growth factor system and somatostatin associated with seawater adaptation of euryhaline fish	Sheridan, Mark
14:15	Prolactin signaling in the highly osmotolerant Mozambique tilapia, <i>Oreochromis mossambicus</i>	Seale, Andre
14:30	Life-history divergence in renal glomerular morphology of Alewife (<i>Alosa pseudoharengus</i>)	Schultz, Eric T.
14:45	Leftovers from fish physiology in osmoregulation: a mechanism of the primary environmental cue inducing drinking in seawater, revisited	Sakamoto, Tatsuya
15:00	Ontogeny of salinity tolerance and ionoregulatory plasticity in mummichog (<i>Fundulus heteroclitus</i>)	Berthelot, Evan
15:15	Functional classification of gill ionocytes in Pacific and Atlantic salmonids	Hiroi, Junya
15:30 - 1600	BREAK	
16:00	There and back again ... almost: tales of an anadromous fish researcher	Shrimpton, Mark
16:15	Ready and swimming: ontogenetic changes preceding emigration in juvenile Alewife (<i>Alosa pseudoharengus</i>)	Schultz, Eric T.
16:30	The genomic underpinnings of adaptation to freshwater in multiple, independently derived populations of landlocked alewife	Velotta, Jonathan
16:45	An increased thyroid hormone apparatus in the brain of chum salmon during smoltification	Wong, Marty Kwok-Shing

17:00	Transcriptomic remodeling of gill ion transport and respiratory mechanisms during metamorphosis in sea lamprey	Shaughnessy, Ciaran
17:15	END	

Date: Thursday July 16	Room: IRC 6	
Time	Title	Author(s)
	TROPICAL Tropical fish: A tribute to Dave Randall	Adalberto Val and Vera Almeida-Val
9:00	Life at the Edge: David Randall's Alpha Helix Legacy in the Amazon	Val, Adalberto
9:15	The tambaqui (<i>Colossoma macropomum</i>), a model neotropical species in fish physiology and toxicology	Wood, Chris
9:30	Variability and diversity: key factors in freshwater fish adaptation	Almeida-Val, Vera Maria Fonseca de
9:45	Current knowledge of lunar synchronized spawning in fish	Fukunaga, Kodai
10:00	External factors influencing air breathing behaviour of a facultative, gut air breathing fish, <i>Panaque maccus</i>	Nelson, Jay
10:15	Tracking tetra transport: how do conditions change for a popular ornamental fish in Brazil?	Addison Marcos, Alexander
10:30 – 11:00	BREAK	
11:00	Microplastic contamination in fish digestive tracts from the Rio Negro and Rio Solimões during an extreme low-water period	Tuvikene, Arvo
11:15	Altered temperature and pH modify key biomarker responses to gasoline exposure in a Neotropical fish	Dal Pont, Giorgi
11:30	Climate-change stress reshapes mRNA-miRNA regulatory networks in the Amazonian fish <i>Apistogramma agassizii</i>	Heinrichs-Caldas, Waldir
11:45	Developing thermal death time models from CTmax data to assess thermal tolerance in <i>Colossoma macropomum</i>	Moura, Lucas
12:00	Behavioural and physiological selection in a tropical ornamental fishery: Evidence from Amazonian species	Pineda, Mar
12:15	Physiological plasticity of tropical coral reef fishes during an extreme marine heatwave on the Great Barrier Reef	Rummer, Jodie
12:30 – 14:00	LUNCH	
14:00	It's O ² K ⁺ - a role for potassium in ion regulation for fish living in the acidic waters of the Rio Negro.	Goss, Greg
14:15	The air breathing physiology and temperature and hypoxia tolerance of the oriental weatherloach (<i>Misgurnus anguillicaudatus</i>)	Burggren, Warren
14:30	Rapid transformation in respiratory performance of coral reef fish larvae settling on a reef	Nilsson, Göran
14:45	Aerial and aquatic oxygen uptake during recovery from exercise and air exposure in <i>Arapaima gigas</i>	Pelster, Bernd

15:00	Chronic climate stress drives gill remodeling and energetic demand in Amazonian fish	Braz-Mota, Susana
15:15	Overcoming DNA degradation in passive eDNA sampling for ichthyofauna monitoring through optimized filter selection and short-fragment markers	Dal Pont, Giorgi
15:30 - 1600	BREAK	
16:00	Validation of environmental DNA metabarcoding for seasonal fish monitoring in the São Salvador reservoir: Identifying hidden biodiversity in the Tocantins River, Brazil	Dal Pont, Giorgi
16:15	Is there reversibility in the expression of microRNAs in <i>Colossoma macropomum</i> exposed to an extreme climate change scenario?	Sanches, Carlos Eduardo da Silva
16:30	END	

Date: Thursday July 16	Room: IRC 1	
Time	Title	Author(s)
	SENSING Sensing the environment: Altered waters and changing behaviours	Tyler Buchinger, Andy Dittman and Keith Tierney
9:00	Tuning out the noise: the role of habituation to underwater sound in freshwater fishes	Beneteau, Gabrielle
9:15	Polystyrene nanoplastics damage lateral-line neuromasts and impair flow-guided behaviors across zebrafish life stages	Gong, Chenxin
9:30	Does hypoxia impair vision in coral reef fish?	Morgan, Rachael
9:45	Cholinergic mucocytes evaluated as candidates for O ₂ sensors in rainbow trout gills	Brink, Deidre
10:00	Do group dynamics shape how fish sense and respond to warming temperatures?	Mercer, Jacob
10:15	Transient loss of olfaction in brown trout (<i>Salmo trutta</i>) following exposure to dissolved organic carbon: Behavioural, physiological, and cellular responses	Zink, Lauren
10:30 – 11:00	BREAK	
11:00	Escaping from danger: Neural mechanisms underlying olfactory aversive behaviours in zebrafish	Yoshihara, Yoshihiro
11:15	Pheromone behavioral antagonists reduce invasive sea lamprey spawning activity in a natural stream during 24-hour trials	Scott, Anne
11:30	Neural and sensory bases of social behavior in African cichlids: from mate choice to parental care	Li, Cheng-Yu
11:45	Pheromonal communication mediating courtship behaviour in zebrafish (<i>Danio rerio</i>) is impaired by environmentally relevant total ammonia nitrogen (TAN) concentrations	Rahman, Emir

12:00	Mechanisms of Neuropeptide Y (NPY) mediated modulation of olfactory sensory neurons in fed and fasted lake sturgeon (<i>Acipenser fulvescens</i>)	Edwards, Tyler
12:15	Impacts of early rearing substrate on the olfactory system development and function in chum salmon (<i>Oncorhynchus keta</i>)	Sevova, Ralitza
12:30 – 14:00	LUNCH	
14:00	Ocean acidification disrupts olfactory signal transduction and receptor structure in the large yellow croaker	Shi, Wei
14:15	Evidence of wildfire-driven differential gene expression in juvenile steelhead olfactory epithelium	Abbott, Morgan M.
14:30	Behavioral responses of cisco (<i>Coregonus artedii</i>) to putative imprinted odors and conspecific odors	Gill, Hannah
14:45	Transcriptional responses of the olfactory system of Pacific salmon to aquatic acidification	Bouyoucos, Ian
15:00	Impact of CO ₂ on Pacific salmon olfaction during key imprinting periods	Reich, Eric
15:30 - 1600	BREAK	
16:00	Fixing an absence of evidence in chemosensory landscaping	Tierney, Keith
16:15	Changes in odorant receptor gene expression in naturally migrating adult Chinook salmon exposed to amino acid odorants as embryos.	Dittman, Andrew
16:30	END	

Date: Thursday July 16	Room: IRC 4	
Time	Title	Author(s)
	CARDIO Cardiac responses to changing environmental conditions	Todd Gillis and Erika Eliason
9:00	Myocardial contraction kinetics and performance underlie differences in cardiac pumping and metabolic capacity between tropical fish species	Gamperl, Anthony
9:15	Thermal stress and Lake Sturgeon cardiac cellular function: an electrophysiological study.	Folkerts, Erik
9:30	Embryonic temperature leaves lasting effects on cardiac phenotype in Atlantic salmon (<i>Salmo salar</i>)	Johansen, Ida
9:45	The role of cardiac remodelling in the acclimation of embryonic and juvenile Sheepshead minnow (<i>Cyprinodon variegatus</i>) to sub-optimal temperatures during embryonic development.	Harman, Adam
10:00	The molecular regulation of cardiac remodeling in trout (<i>Oncorhynchus mykiss</i>) caused by thermal acclimation and/or a high salt diet	Robinson, Sean

10:15	Structural cardiac plasticity in Atlantic bluefin tuna (<i>Thunnus thynnus</i>) under aquaculture and thermal regimes	Wierer, Amelie
10:30 – 11:00	BREAK	
11:00	Seasonal patterns in the cardiac temperature sensitivity of northern fishes	Williams, Emily
11:15	Egg size impacts embryonic cardiac thermal performance and growth in Atlantic salmon (<i>Salmo salar</i>)	Morefield, Robert
11:30	Chronic stress induced cardiac remodeling, depleted energy reserves, and reduced swimming performance of Atlantic Salmon	Opinion, April Grace
11:45	Acclimation to mild static and cyclic hypoxia reduces cardiac thermal tolerance in brook char (<i>Salvelinus fontinalis</i>); potential contribution of novel lactoylated amino acids	MacCormack, Tyson
12:00	Anaerobic metabolism and acidosis: supporting heart function in anoxic pacific hagfish (<i>Eptatretus stoutii</i>)	Yutsyschyna, Maria
12:15	Swimming performance of hagfish in hypoxia	Shaftoe, Jared
12:30 – 14:00	LUNCH	
	DEEP BLUE The deep blue sea: Probing the physiology and behaviour of large marine fishes	David McKenzie and Holly Shiels
14:00	Tracking the physiology pelagic fish from the lab to the blue Serengeti	Block, Barbara
14:15	(Continued)	Block, Barbara
14:30	Natural isotope tracer-based estimation of field metabolic rates of wild pelagic fishes: approaches, applications and challenges	Trueman, Clive
14:45	(Continued)	Trueman, Clive
15:00	Effects of body size on spontaneous swimming activity and energetics in Atlantic bluefin tuna	Thambithurai, Davide
15:15	Atlantic bluefin tuna recovery from catch-and-release	Birnie-Gauvin, Kim
15:30 - 1600	BREAK	
16:00	Mesothermic fishes face high fuel demands and overheating risk in warming oceans	Payne, Nicholas
16:15	Fine-scale spawning dynamics of Atlantic bluefin tuna	Cortese, Daphne
16:30	Effects of seasonal temperatures on cardiac and swimming activity in Atlantic bluefin tuna	McKenzie, David J.
16:45	Body size relationships in cardiac morphology and electrocardiographic intervals of Atlantic Bluefin tuna	Bjarnason, Asgeir
17:00	Thermal sensitivity of cardiac function in the Greenland Shark	Shiels, Holly
17:15	(Continued)	Shiels, Holly
17:30	END	

Date: Thursday July 16	Room: IRC 5	
Time	Title	Author(s)
	DIGESTION The fish gut: Digestion and beyond	Patricia Ferreira and Jonathan Wilson
9:00	Genomic foundations of the vertebrate stomach	Castro, Filipe
9:15	(Continued)	Castro, Filipe
9:30	Food for thought: The role of serotonin in mediating energy metabolism via the gut	Gomes Ferreira, Patricia
9:45	Structural support or functional necessity: Evaluating the importance of the gastric proton pump β -subunit in gastric function using gene knockout.	Seguin, Aiden
10:00	Thermal plasticity of intestinal function in satiated and starved European seabass	Ruhr, Ilan
10:15	Convergent gene losses and pseudogenizations in stomachless fishes	Kato, Akira
10:30 – 11:00	BREAK	
11:00	Digestive enzyme activities in prickleback fishes (Stichaeidae) appear to be driven by gene expression	German, Donovan
11:15	Integrating transcriptomics, qPCR and electrophysiology to assess intestinal adaptation to plant-based diets in Greater Amberjack	Fuentes, Juan
11:30	Secondary bile acids impact rainbow trout homeostasis through modulation of feeding and metabolism	Soengas, José L
11:45	Gastrointestinal nitrogen metabolism in a voracious predator: studying one or studying millions?	Wood, Chris
12:00	Multigenerational gut-brain axis remodeling in marine medaka under chronic hypercapnia	Liu, Tzu-Yen
12:15	Specific dynamic action and other related inconveniences	Goodrich, Harriet
12:30 – 14:00	LUNCH	
	INVASIVE The physiology, ecology and control of invasive fish species	Mike Wilkie and Pedro Guerreiro
14:00	A rainbow trout's guide to recovering from sea lamprey parasitism	Alderman, Sarah
14:15	A potential strategy to offset the greater tolerance of invasive sea lamprey to the lampricide, TFM, in warmer waters	Zhao, Zhijun (Tim)
14:30	Can rock ramp fishways exclude juvenile Common Carp while allowing passage of native fishes?	Myrick, Christopher
14:45	Fine scale response of invasive carp and native fish to an acoustic deterrent at a large riverine lock	Albers, Janice
15:00	Redds, carcasses, and contaminants: ecological impacts of invasive Chinook salmon in South American rivers	Fierro, Pablo
15:15	Born to survive: Movement ecology of invasive round goby during early life stages	Zarini, Sina

15:30 - 1600	BREAK	
16:00	Can biotic resistance to invasions come from behavioural competition? – a case study of gobies from the north east Atlantic	Green, Leon
16:15	Thermal plasticity with physiological trade-offs in the invasive cichlid <i>Australoheros facetus</i> under warming scenarios in Mediterranean-type rivers	Guerreiro, Pedro M
16:30	END	

Date: Thursday July 16	Room: IRC 3	
Time	Title	Author(s)
	MECHANISMS Bridging experimental biology and conservation: From mechanism to application	Heather Bauer Reid, Vincent Mélançon, Sandra Ann Binning, Mar Pineda and Leon Tran
9:00	Oxygen limitation decreases the recovery potential of exploited fish populations; the case of Baltic cod	Behrens, Jane
9:15	(Continued)	Behrens, Jane
9:30	Developmental acclimation to local thermal environments shapes physiological and hormonal warming responses in anemone-associated reef fish	Bauer Reid, Heather
9:45	Projecting habitat suitability for the convict tang (<i>Acanthurus triostegus</i>) in Hawai'i under future ocean conditions using physiology-based models	Tran, Leon
10:00	Seasonal thermal tolerance, growth and acclimation capacity in newly settled European plaice (<i>Pleuronectes platessa</i>) from Swedish coastal nursery habitats	Tamarit Castro, Elena
10:15	Impact of food restriction on the thermal tolerance and recovery rate of a differently acclimated endangered freshwater fish	Levet, Marie
10:30 – 11:00	BREAK	
11:00	Microplastics in riverine fishes – from the river to the lab – and back to the river	Shiels, Holly
11:15	(Continued)	Shiels, Holly
11:30	Acute shifts in water chemistry during stocking influence nutrient absorption by young-of-year lake sturgeon	Genz, Janet
11:45	Endurance swimming tests to inform entrainment risk	Hannan, Kelly
12:00	Physiology, movement, and ecology, oh my! Applying scientific insights to monitor fish at Site C Dam	Pleizier, Naomi
12:15	Age is just a number? Exploring the effects of establishment times on fish stress in marine protected areas.	Beach, Riley
12:30 – 14:00	LUNCH	

14:00	Bridging physiology and conservation science to promote the recovery of at risk salmonids	Enders, Eva
14:15	(Continued)	Enders, Eva
14:30	Molecular detections of food deprivation in juvenile chinook salmon: from lab development to field application	Bugg, William
14:45	Tracking road salt contamination through community monitoring: Surface water chloride trends in Vancouver streams, and the implications for local salmonids.	Kilgour, Clare
15:00	Thermal adaptation in freshwater stream fishes	Stanton, Olivia
15:15	Performance, plasticity, and resilience to warming in diverse populations of brook trout	Stewart, Erin
15:30 - 1600	BREAK	
16:00	Acclimation temperature reverses warming-rate effects on upper thermal limits in zebrafish	Ern, Rasmus
16:15	Investigating transcriptomic signatures to isolate a cold shock signal from impingement stress in yellow perch	Thapar, Myra
16:30	Minimally invasive sampling for transcriptional profiling to monitor acute stress and recovery in rainbow trout	Jeffries, Ken
16:45	Leveraging empirical science, agent-based modeling, and model sensitivity testing to make informed management decisions that support fish migration in data-poor systems	Dressler, Terra
17:00	Twenty years on ice: early-life performance following long-term milt cryopreservation in rainbow trout	Kostelnik, Mageleate
17:15	END	